Particle-Origin Primitives

A ULD/UNS Confinement-Geometry Account of Free Modifiers and Indo-European Bound Exponents

2026-02-03

Maks Panner,

makspanner@gmail.com

Abstract

This article develops a single internal claim that recurs across the Morphodynamics program: the primitive inventory that drives grammatical evolution is particle-stratified. Free or weakly bound discourse / clausal modifiers and certain "primitive" bound exponents (illustrated by Indo-European suffixal atoms such as -s, -t, and -m̥) are not clausal or syntactic primitives. They are particle-origin elements whose apparent clausal behavior emerges from confinement geometry. The analysis is written entirely in the native morphodynamic formalism: a ghost-slot ("well") is a directed confinement geometry with (i) a state inventory V, (ii) a licensed transition set E, (iii) depth/weight w, and (iv) a distributional wavefunction ψ that supports canonical wave-dynamic outcomes (drift, diffusion, fusion, absorption, elimination, collapse). Within this geometry, particle material occupies the high-optionality surface and transitional channels, and may stabilize into bound morphology via absorption and fusion. The paper formalizes particle-origin diagnostics with theorem-style statements (including a Particle Origin Lemma for -s and its extension to the compact suffix set), supplies well-internal predictions, and provides a battery of schematic but operational figures, tables, and adjudication templates that can be populated with the manuscript's measurement program.

Keywords

Morphodynamics; ULD/UNS; confinement geometry; ghost-slot; particle stratification; MORPHIX; Indo-European; -s; -t; -m̥; wave dynamics; directed graph well; typed transitions.

1. Problem statement and scope

A recurring empirical observation in the article set is that “functional” inventories – discourse modifiers, clause-level operators, pronominal stems, and certain apparently minimal suffixes – behave unlike lexical roots. They are small in number, highly recurrent, cross-categorial in their distribution, and unusually stable under diachrony. The morphodynamics account does not treat these behaviors as special exceptions. Instead, it treats them as the expected outcome of a particle-stratified confinement geometry.

The target claim of this article is deliberately narrow and internal: free or weakly bound discourse / clausal modifiers, and the primitive bound morphology traditionally treated as the smallest grammatical atoms (e.g., Indo-European -s, -t, -m̥), are not clausal or syntactic primitives. **They are particle-origin elements**. Their “clausal” effects are emergent: they arise when particle material is confined in a ghost-slot and pushed along a binding trajectory that culminates in suffixal stabilization.

The paper is add-only with respect to the previous works. It does not import external ontologies. The only notions employed are those explicitly present in the morphodynamics well formalism, the morphix feature-geometry program, and the proof sheet defining particle status for -s (and allied exponents).

The remainder is structured as follows: §2 states the confinement-geometry primitives; §3 defines particle vs root primitives and the particle stream; §4 states canonical wave-dynamic outcomes and signatures; §5 introduces morphix as a particle inventory and states its criteria; §6 provides theorem-style results for particle-origin bound exponents; §7 models pathways from free particles to bound morphology inside the well; §8 lists predictions, falsifiers, and measurement templates; §9 concludes.

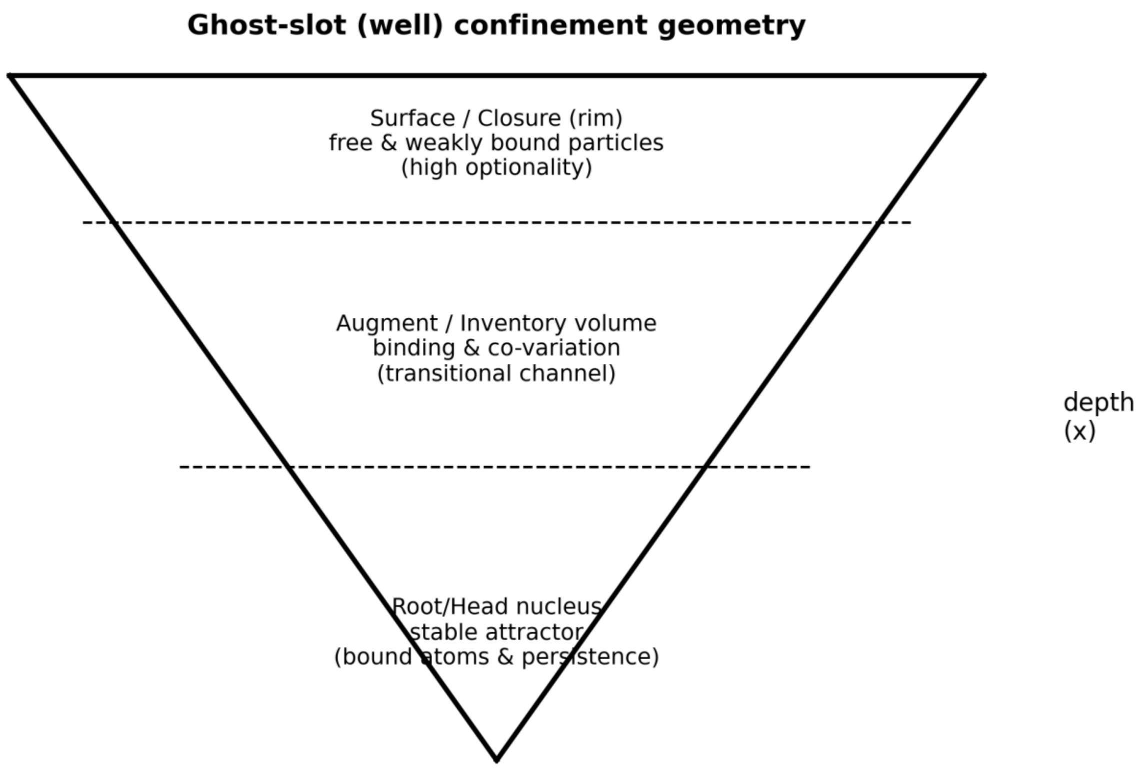


Figure 1 — Ghost-slot (well) as confinement geometry with three functional zones (schematic).

More precisely: the same well geometry that models vowel nuclei and morphological basins also models functional inventories. The difference is not a change in geometry, but a change in where the distribution concentrates: particle material lives in the high-permeability surface and transitional channels, whereas root-anchored material occupies deep stable basins. This article develops that claim for discourse modifiers and Indo-European suffixal atoms.

2. The ghost-slot (well) as a universal confinement geometry

In the morphodynamics architecture, a "well" (also called a ghost-slot) is an abstract confinement geometry induced over a restricted subsystem. It is not a physical well and it is not tied to any particular representational layer by definition. The well is a geometric object characterized by a stable internal construction: vertices, admissible transitions, depth/weights, a distributional wavefunction, and regime thresholds.

The point of the well is to make a single claim computable: once a subsystem is restricted (by inventory, by locality, by paradigm, by orthography, by any admissible gating condition), its distribution behaves as an object ψ over the well. The well's parameters then impose constraints on how that distribution can move, broaden, merge, be captured, vanish, or collapse.

For convenience, the internal construction is restated in a primitive table. The linguistic interpretation column is intentionally generic: it is read in ULD/UNS (Universal Linguistic Domain/Universal Noun Slot) terms as an abstract geometry of confinement and transitions, independent of a particular projection.

2.1 Primitive parameter table for well construction

Primitive	Formal role	ULD/UNS interpretation	Empirical handle (within program)
Inventory V	Vertex set of states	Attractor states in a confinement geometry; stable nuclei / stable basins	Frequency mass; distributional anchoring
Edge set E	Admissible directed transitions (ordered	Licensed state→state movement; permitted	Inventory of attested transitions; gaps

	pairs)	alternations; compositional transitions	
Depth / weight w	Cost / stability parameter (potential component)	Confinement strength; resistance to leaving a basin; stability gradient	Conditional distributions; relative persistence under perturbation
Wavefunction ψ	State distribution over forms	Amplitude-like object encoding morphological mass allocation	Token counts; normalized probabilities
Eigenstates	Stratified stable modes	Stable distributional strata; plateaus; quasi-register modes	Clustered distributions; plateau behavior
Trajectory τ	Typed transition object	Directed change within a single slot; licensed internal motion	Dynamic signatures; path statistics
Field Φ(·)	Contextual coupling	Cross-locus interaction; paradigm/context coupling in the ambient ULD	Co-variation matrices; conditional shifts
Force F	Vector pressure on ψ	Directional pressures producing drift/restoration/repulsion	Asymmetric change; repair directions
Threshold / regime	Phase boundary; instability gate	Onset of turbulence/collapse; renewal boundary in the geometry	Variance spikes; entropy and permeability changes

The table is the only required content to define a well abstractly. Everything else—pronoun systems, bound suffixes, clausal modifiers—is a specialization of how ψ concentrates and how E and w constrain τ-licensing.

3. Primitive inventory: particle material vs root material

The particle-stratification claim requires a primitive distinction. The proof sheet for -s makes this explicit by introducing a particle predicate Particle_L that is defined in terms of two properties: (i) relational expression (RelExp_L) and (ii) failure of root status (¬Root_L). The same sheet emphasizes that bound exponents may be surface-mobile but structurally non-clausal: they behave as particle-origin elements that later stabilize into bound morphology.

In the morphodynamic reading, "particle" is not "small word." It is a confinement type. Particle elements are those whose distribution is primarily licensed by relational or connective environments rather than by root anchoring. They therefore concentrate at the surface and transitional channels of the well, with high boundary permeability and high optionality. "Root" elements are those whose distribution is anchored by stable basins and persistent paradigmatic commitments; they concentrate in deeper stable zones.

The same logic is used in the MORPHIX program: pronominal and adverbial stems are treated as a compact closed-class inventory derived from an older particle layer, and explicitly contrasted with lexical roots. The particle layer is non-derivable, non-compounding, and non-ablauting; it is "atomized" material from which larger systems are built by attachment and fusion.

3.1 Formal predicates (particle vs root)

> Definition (Root predicate). Root_L(x) is true of x iff x is a lexical root object in the relevant language-internal ontology (root-anchored, open-class, able to host derivational structure as head).
>
> Definition (Relational expression). RelExp_L(x) is true of x iff x is licensed primarily as a relational exponent (connective / deictic / indexical / clausal-modifying function), not as a root.

Environment (OBJ)	**What is linked (Φ-coupling locus)**	**Expected signature in well terms**	**Fast falsifier (clausal-primitive reading)**
Discourse adverbial / stance modifier	links clause to discourse state (context lattice)	high boundary permeability; broad context spread (Diffusion) without	requires fixed clausal spine position; cannot occur as free/weak particle across hosts

		root anchoring	
Clause-edge linker (complement / subordinator-like)	links proposition to embedding relation (typed edge in E)	edge-local transitions; co-variation with neighboring connective inventory (Φ shifts)	behaves like mandatory syntactic head with strict selection independent of particle inventory
Degree/scale operator site	links predicate scale to comparison frame (Φ over degree lattice)	drift in centroid over degree contexts; moderate entropy change (Drift)	scope only derivable via operator movement; particle distribution cannot simulate it
Anaphoric/deictic resolver site	links token to accessibility set (reference channel, not root nucleus)	stabilization can increase confinement (w) while type remains Particle_L	forces the form itself to be a root-like referential nucleus
Coordination / indexing interface	links parallel loci (pairing / alignment)	diffusion across coordination hosts; overlap increase with neighboring particles	cannot occur outside coordination; no free particle alternants
Repair/contrast interface (metalinguistic)	links evaluation layer to utterance content	collapse-like discontinuities under perturbation; non-linear response	only syntactic reconstruction explains repair; particle dynamics cannot

Licensing inventory for RelExp_L (object-language environments)

The inventory below lists the environments treated as relational/particle-licensed (RelExp_L) in this paper. They are modeled as Φ-coupling sites (cross-locus linkage) for Particle_L material, not as clausal primitives.

> Definition (Particle predicate). Particle_L(x) ≡ RelExp_L(x) ∧ ¬Root_L(x).
>
> Particle_Stream_L is the set of x such that Particle_L(x). The particle stream is the pool from which both free modifiers and bound exponents may be drawn, depending on the well's depth regime and the binding trajectory.

3.2 The particle origin claim for bound exponents

The crucial point is that being bound does not imply being clausal or syntactic primitive. In the morphodynamic ontology, binding is a depth effect: a particle can be progressively confined and reweighted until it behaves as a stable suffixal atom in the deep region. What changes is not its primitive type, but its position in the confinement geometry and the stability of its basin. This observation is used to re-read "primitive" Indo-European suffixes (-s, -t, -m̥, and other compact suffixal atoms). Their structural minimality is not evidence of clause-primitive status; it is evidence of particle-origin stabilization and basin capture.

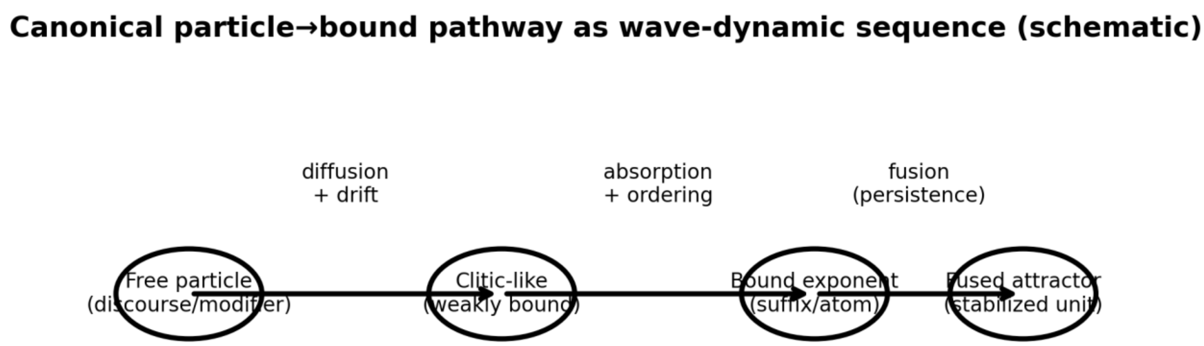


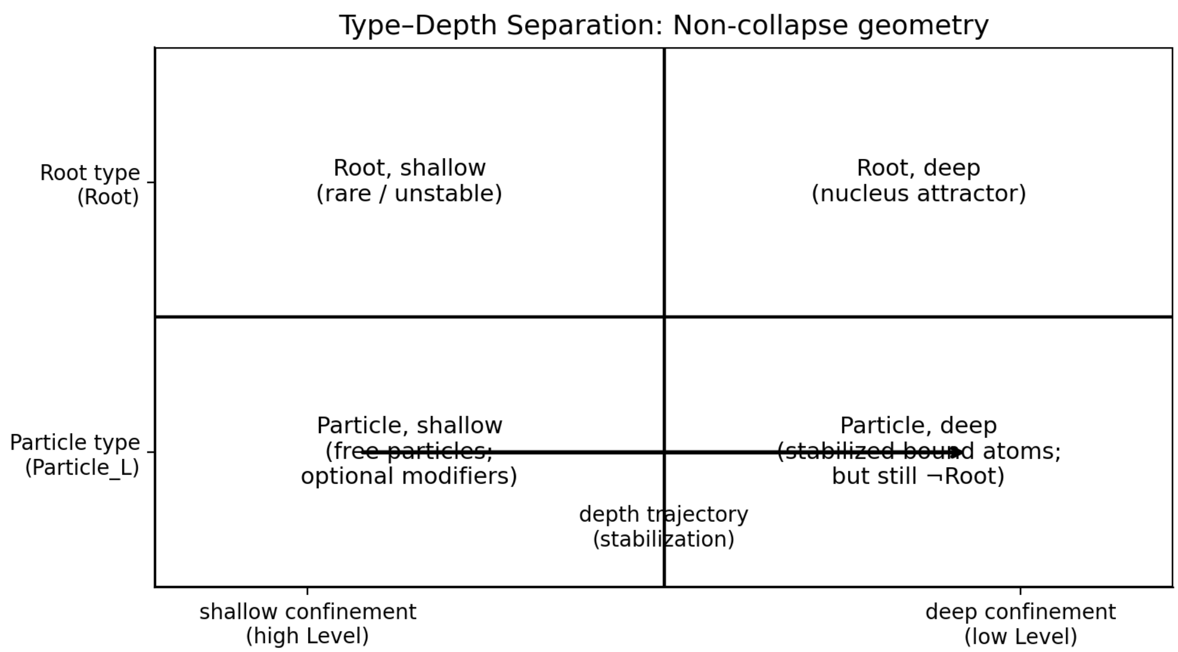
Figure 2 — Particle→clitic→suffix stabilization as a wave-dynamic trajectory inside the well (schematic).

Axiom NC (Non-collapse): TYPE □ DEPTH
NC1. TYPE = {Root, Particle_L} is an anchoring predicate; DEPTH = w / V(x) is a confinement coordinate.
NC2. DEPTH measures stability/boundness trajectories; it does not retype a Particle_L item into a Root.
NC3. Particle_L material may increase confinement (stabilize, bind, fuse) while remaining ¬Root.
NC4. Therefore: deep confinement ≠ root anchoring; pronoun/atom stabilization does not entail root status.

3.3 Type–Depth grid and the Non-collapse axiom (visual discipline)

To prevent category slippage ("deep" ⇒ "root"), the model enforces an explicit separation between TYPE and DEPTH.

Figure 2A — Type–Depth grid: Root vs Particle_L crossed with shallow↔deep confinement (schematic).



This grid makes explicit that depth trajectories (e.g., particle→clitic→suffix stabilization) are compatible with Particle_L type. In particular, a stabilized particle inventory can occupy deeper basins in the well without collapsing into root-anchored argument structure.

3.4 Falsifiers at a glance (constraints-first)

The argument is constrained by falsifiers. If any of the following holds, the particle-origin claim fails (full detail remains in §8.4).

- F1. A compact suffix atom behaves as a root: it is open-class, head-capable, supports derivational expansion as a head, and participates freely in compounding. This would satisfy Root_L and contradict Particle_L.

- F2. A compact atom's distribution is insensitive to relational environments (fails RelExp_L) and is instead anchored as a lexical nucleus. This would contradict the relational licensing assumption.
- F3. Diachronic stabilization of the atom does not show the expected wave-dynamic signatures (no diffusion during spread; no absorption/fusion during binding; no curvature changes). This would undermine the morphodynamic pathway explanation and would require a different internal mechanism.

4. Wave dynamics: canonical outcomes and diagnostic indices

Wave dynamics supplies the time axis: once ψ is treated as an object on the well, change types that are conflated in descriptive work separate into distinct dynamical outcomes. The manuscript's canonical list is: drift, diffusion, fusion, absorption, elimination, and collapse. Each outcome is intended to have a distinct diagnostic signature in indices such as entropy, overlap, boundary permeability, and distributional curvature.

For the present problem, the importance of wave dynamics is methodological. The claim that "bound morphology is particle-origin" becomes testable only if the pathway from free particle to bound exponent has a recognizable dynamical signature. For example, the transition from a free discourse modifier to a suffixal atom should show diffusion (entropy increase; permeability increase) followed by absorption or fusion (asymmetric capture; peak count reduction; stabilized overlap).

4.1 Canonical wave-dynamic outcomes

Dynamic outcome	Morphophysical description	Expected signature	Typical correlates (internal reading)
Drift	mass shifts within or across basins	gradual centroid change; moderate entropy change	frequency redistribution; gradual alternation bias
Diffusion	broadening / leakage across barriers	entropy increase; boundary	context generalization;

		permeability increase	analogical spread
Fusion	two peaks coalesce into one basin	peak count decrease; overlap increase then stabilizes	merger; reanalysis; atomization of compound
Absorption	deep basin captures shallow basin	asymmetric loss; increased attraction signature	dominant form overrides competitor; capture by deeper basin
Elimination	peak disappears without new basin	mass relocation; reduced variance after removal	loss of rare exponent; leveling
Collapse	rapid global reconfiguration	spike in instability indices; discontinuity	system reset; rapid restructuring

The table is used later (§7) to map particle→bound trajectories to outcome signatures.

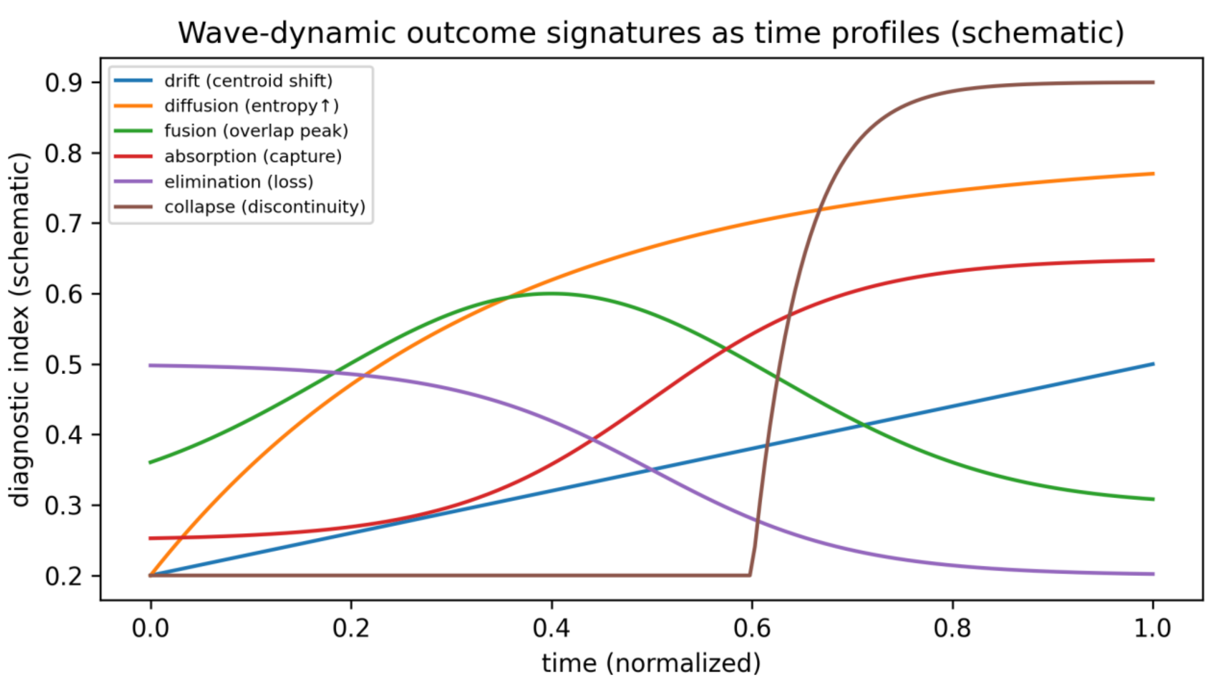


Figure 3 — Schematic time profiles for the diagnostic indices associated with each wave-dynamic outcome.

5. MORPHIX: particle stratification of pronominal and adverbial morphology

The Morphix program explicitly frames its object of study as particle-layer material. It argues that PIE pronoun stems do not behave like lexical roots: they are small, closed-class, non-derivable, and do not participate in the full combinatorics of root morphology. Instead they reflect an older particle layer, from which pronoun systems and adverbial systems are assembled by attachment and fusion. This directly supports the present article's premise: what looks "clausal" is often particle-origin.

In Morphix program, the primitive inventory is organized into three strata: a particle layer (source of Morphix stems), a root/stem layer, and a derivational/inflectional layer. The particle layer includes deictic and relational stems, plus a compact set of suffixal atoms that participate in bound morphology without being treated as root material.

The purpose of importing Morphix here is not to redo its entire analysis, but to isolate what it contributes to the well formalism: (i) a compact particle inventory that is stable and portable across constructions, (ii) a feature-geometry scaffold for partitioning that inventory, and (iii) explicit criteria distinguishing particle atoms from lexical roots.

5.1 MORPHIX criteria as particle diagnostics

MORPHIX defines its stem inventory with criteria intended to isolate atoms. The most relevant for the particle-origin thesis are: closed-class behavior (small inventory), lack of derivational expansion as head, lack of compounding, and restricted ablaut behavior. These criteria align with Particle_L: the stems are relational expressions and are not roots.

For later measurement, the criteria can be re-expressed as observable constraints on ψ and E: closed-class size constrains $|V|$; lack of compounding restricts E to a small set of state-to-state transitions; restricted ablaut restricts allowed alternation edges; and high recurrence constrains ψ to have strong peaks.

5.2 Stratified inventories: particle layer, roots, derivational/inflectional layer

Stratum	Role in well	Characteristic properties	Representative content (from manuscripts)

Particle layer	Surface + transitional channels	closed class; relational; high permeability; optionality	MORPHIX stems; deictic/relational particles; free discourse modifiers
Root/stem layer	Deep basins (host attractors)	open class; head-capable; derivational hosting	lexical roots; stable nuclei for attachment
Derivation/Inflection layer	Deep stabilization + ordering constraints	structured paradigms; ordering; persistence	bound exponents; compact suffix atoms; paradigmatic affixes

Importantly, the particle layer and the derivation/inflection layer are not the same: particle material can be free or weakly bound, but it can also stabilize into bound atoms. Boundness is a depth effect; particle status is a primitive type effect.

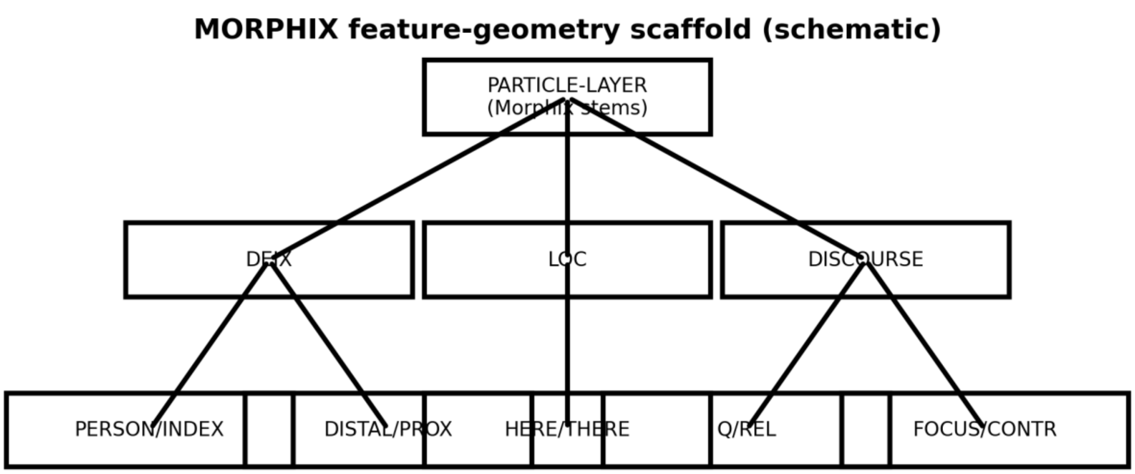


Figure 4 — MORPHIX feature-geometry scaffold for partitioning particle-layer stems (schematic).

5.3 The compact suffix set as particle-origin atoms

MORPHIX explicitly lists a compact set of suffixal atoms (a small set of exponents) and interprets them as part of the particle-layer system that stabilizes into bound morphology. The current article treats this compact set as the canonical testbed for particle-origin bound exponents. In particular, the proof sheet demonstrates that -s is particle origin; and MORPHIX places -s together with other minimal atoms (including -t and -m) as part of a compact exponent inventory used in pronominal and adverbial morphology.

5.4 Compact suffix set (illustrative inventory)

Suffix atom	Status (primitive type)	Distributional role	Well interpretation
-s	Particle-origin (RelExp ∧ ¬Root)	widespread bound exponent; pronominal/adverbial morphology	particle atom stabilized by capture/fusion
-t	Particle-origin (parallel diagnostic)	minimal bound exponent in compact set	particle atom stabilized by capture/fusion
-m / -m̥	Particle-origin (parallel diagnostic)	minimal bound exponent; deictic/indexical roles in system	particle atom stabilized by capture/fusion
-r	Compact atom (as listed in MORPHIX)	bound exponent participating in pronominal/adverbial morphology	deep basin marker for particle atom
-d	Compact atom (as listed in MORPHIX)	bound exponent; contrast/series marking in system	attractor-shaping exponent
-o	Compact atom (as	stem/ending atom	boundary marker /

	listed in MORPHIX)	used in pronominal stems	closure of particle stem

6. Formal results: particle-origin bound morphology is not clausal primitive

This section packages the particle-origin claim in theorem form, using only predicates and commitments that occur in the previous works. The goal is not to "prove language facts" by logic alone; it is to make explicit what follows once the particle predicates and the confinement geometry are assumed.

Notation. Let L be a language-internal system; let V be the well's state inventory; and let $S \subset V$ be the compact suffix set treated as atomized exponents in the manuscripts. The particle predicate is Particle_L(x) ≡ RelExp_L(x) ∧ ¬Root_L(x).

Theorem 1 (Particle Origin Lemma for -s)

If RelExp_L(-s) and ¬Root_L(-s), then Particle_L(-s).

Proof. Immediate by the definition of Particle_L(x) as RelExp_L(x) ∧ ¬Root_L(x). The substantive work is empirical: establishing that -s behaves as a relational exponent and fails root status in the manuscript's ontology. The proof sheet for -s is exactly that empirical establishment, and it notes the extension to other minimal exponents (-t and -m̥) as belonging to the same primitive class.

Theorem 2 (Compact suffix atoms are particle-origin under the same predicate)

Let $a \in S$. If RelExp_L(a) and ¬Root_L(a), then Particle_L(a).

Proof. Same as Theorem 1: Particle_L(a) is definitional. The value is that it cleanly separates a type claim from a depth claim. The type claim is: a is particle-origin if it satisfies the particle predicate. The depth claim is: a is bound if its ψ-mass is confined in a deep basin and its exit probability across barriers is low (high w). The manuscripts' program is to show that the compact atoms behave as particle types, while also showing that some of them stabilize into deep bound basins.

Proposition 1 (Non-clausal primitive consequence)

> If Particle_L(a), then a is not a clausal/syntactic primitive in the ontology; any clausal effects of a arise from confinement (Φ coupling and τ-licensing) rather than from a primitive clause-operator status.

Justification. This proposition is an ontological stipulation in the manuscripts: particle material is relational and connective, and it is explicitly not treated as the primitive clause skeleton. The clause effects are emergent because particle elements interact with the well via Φ coupling (contextual interaction) and τ-licensing (typed transitions) that may be realized at higher levels. In other words, the clause sees the particle as a boundary condition, not as its primitive generator.

Theorem 3 (Particle→bound stabilization is a depth effect)

> For any particle element p with Particle_L(p), there exist well regimes in which p is realized as (i) free/weakly bound surface material or (ii) bound deep morphology, depending on the depth/weight profile w and the coupling field Φ.

Proof sketch. The well formalism distinguishes type from confinement. p remains in V as the same vertex state, but its occupancy mass ψ(p) and the barrier costs w on outgoing edges determine whether p behaves as optional surface material or as a stabilized bound exponent. When permeability is high (low w) and Φ supports multiple contexts, diffusion and drift permit p to spread as a free modifier. Under capture (absorption) and merger (fusion), ψ(p) becomes concentrated and the effective barrier cost for leaving the p-basin increases, yielding bound behavior.

Proposition 4 (Particle-edge locality; non-definitional).

Let ⟨ V,E,w⟩ be a particle-type well with a feature metric δ on V (induced by the MORPHIX feature geometry). Assume Particle_L inventories are compact and insertion/alternation is feature-local. Then admissible transitions concentrate on δ-local neighborhoods: $\forall (u\to v)\in E$, $\delta(u,v) \le k$ for small k, and long-distance alternations arise only as multi-step paths $\tau = (u\to\ldots\to v)$ with intervening barriers.

Proof sketch. In MORPHIX, particle atoms are realized by local feature packages. A single-step alternation $u \to v$ that flips many independent features corresponds to a large barrier in $V(x)$ (multiple constraints violated at once), so its transition weight is suppressed. The well therefore realizes remote changes as compositions of feature-local steps, which preserves the compactness and reusability of the particle inventory while keeping E compatible with the locality constraints.

Falsifier. If a purported Particle_L inventory exhibits dense, high-weight edges between feature-remote states (large δ) as single-step alternations, the locality consequence fails and the analysis must be revised (either the item is not Particle_L, or the feature geometry/metric is wrong).

Theorem 4 (Weighted Diffusion Descent; non-definitional).

Let a well/ghost-slot be a weighted directed graph $G = (V, E, w)$ with vertex set V (states), edge set E (licensed ordered transitions), and weight function w on edges (resistance/cost). Let ψ_t be a probability distribution over V at step t. Define the weighted diffusion operator D_w by a Markov kernel P_w on E:

$$P_w(u \to v) = \exp(-\beta \cdot w(u,v)) / Z(u) \quad \text{for } (u,v) \in E, \quad Z(u)=\sum_{(u,z)\in E} \exp(-\beta \cdot w(u,z));$$
$$\psi_{t+1} = \psi_t \cdot P_w.$$

Assume P_w is irreducible and reversible with stationary distribution π_w (the equilibrium distribution induced by w on G). Define the well free-energy functional as the KL divergence to equilibrium:

$$\mathbb{F}(\psi) := KL(\psi \,||\, \pi_w) = \sum_{v\in V} \psi(v) \cdot \log(\psi(v) / \pi_w(v)).$$

Then along diffusion steps $\psi_{t+1}=\psi_t \cdot P_w$, free energy decreases monotonically:

$$\mathbb{F}(\psi_{t+1}) \leq \mathbb{F}(\psi_t), \text{ with strict inequality unless } \psi_t = \pi_w.$$

Corollary (signature link). When the equilibrium π_w is approximately flat within a restricted domain (or after factoring out π_w), the Shannon entropy $H(\psi_t)$ increases under diffusion until plateau. Therefore Diffusion is not merely named; it is diagnostically bound to (i) entropy increase and (ii) support expansion through E (a permeability effect), with $\beta \cdot w$ controlling the rate and asymmetry.

Proof sketch.

(1) P_w is a Markov operator on the simplex $\Delta(V)$ with stationary π_w by construction. (2) For reversible Markov kernels, KL divergence to stationarity is contractive under the channel $\psi \mapsto \psi \cdot P\_w$ (data-processing inequality for relative entropy / convexity of KL). (3) Contractivity gives $\mathbb{F}(\psi\_\{t+1\}) \leq \mathbb{F}(\psi\_t)$. (4) If π_w is uniform (or if we measure entropy relative to π_w), decreasing KL is equivalent to increasing entropy toward equilibrium. This yields the canonical Diffusion signatures in the wave-outcome matrix.

6.1 A proof-oriented reading of "primitive bound morphology"

A common descriptive assumption is: "if an exponent is tiny and ubiquitous (-s, -t, -m̥), it must be a clause primitive." The theorems above show that, in the manuscript's ontology, this inference is not licensed. Tiny + ubiquitous is compatible with being a particle atom that has become deeply confined. The crucial distinction is between (i) a primitive generator of clause structure and (ii) a primitive atom of a particle stream that can be bound or free depending on the well. Only (i) would count as a "clausal/syntactic primitive," and the manuscripts explicitly assign the compact suffix atoms to (ii).

7. Wave-dynamic classification for particle material and bound exponents

This section turns the theorem statements into diagnostics: given an observed change in a particle inventory or a suffixal atom, which canonical wave-dynamic outcome is implicated, and what signatures should be observed? The key idea is to treat "particle-origin" as a constraint on V and E (type), and to treat "free vs bound" as a property of ψ mass allocation and w barriers (depth).

To keep continuity with the synopsis, we first restate the suffix-centered affixation table used previously, then generalize it to particle systems (discourse modifiers and pronominal stems).

7.1 Affixation (suffix-centered) phenomena mapped to outcomes

Affixation phenomenon (suffix-centered)	Well band (depth)	Canonical outcome(s)	Diagnostic signature(s) (as defined)

Productivity increase (suffix starts attaching to more stems/contexts)	middle → deep	Diffusion (often with some Drift)	Entropy increase; boundary permeability increase (diffusion). Gradual centroid change (drift).
Competition among allomorphs / rival suffixes (one variant slowly gains share)	middle / deep	Drift	Gradual centroid change; moderate entropy change.
One suffix “wins” over another (dominant pattern captures a weaker competitor)	middle / deep	Absorption	Asymmetric loss; increased “attraction” signature (capture by deeper basin).
Resegmentation / coalescence (root+suffix reanalyzed; or two peaks become one)	deep	Fusion	Peak count decrease; overlap increase then stabilizes.
Loss of a marginal suffix (rare affix disappears; leveling removes it)	middle / surface (often)	Elimination	Mass relocation; reduced variance after removal.
System-wide affixal restructuring (many suffixes reweighted/reanalyzed quickly)	cross-band	Collapse	Spike in instability indices; distributional discontinuity (rapid global reconfiguration).

The depth assignments follow the manuscript’s stratification logic: surface optional material is shallow, paradigm ordering is middle, and persistence/fusion is deep. Importantly, particle origin is compatible with any of these depth regimes: particles can be shallow (free), middle (clitic-like), or deep (stabilized atoms).

7.2 Particle-inventory phenomena mapped to outcomes

Discourse and pronominal inventories are particle inventories in the manuscripts: they are closed-class relational material. Their wave dynamics therefore follow the same canonical outcomes, but with different typical correlates: diffusion corresponds to contextual spread of a particle; drift corresponds to gradual preference shifts; absorption corresponds to one particle capturing the contexts of another; fusion corresponds to cliticization or bound-atom formation; elimination corresponds to loss of a particle; collapse corresponds to rapid inventory reset.

Particle-system phenomenon	Well band (depth)	Canonical outcome(s)	Diagnostic signature(s) (as defined)
Free discourse modifier spreads to new clause environments	surface → middle	Diffusion (+ Drift)	Entropy ↑; boundary permeability ↑; gradual centroid shift.
Weak particle becomes enclitic under locality pressure	middle (binding channel)	Absorption / Drift	Asymmetric loss of free variant; attraction signature; centroid shift toward bound host contexts.
Particle coalesces with host into bound exponent	deep stabilization	Fusion	Peak count ↓; overlap ↑ then stabilizes; reduced permeability across boundary.
Two competing particles split contexts then one captures the split	surface / middle	Absorption	Asymmetric loss; attraction signature; entropy may fall after capture.
Rare discourse	surface	Elimination	Mass relocation;

particle disappears without replacement			reduced variance after removal.
Inventory re-weights rapidly under structural perturbation	cross-band	Collapse	Discontinuity; spike in instability indices (entropy, curvature, permeability).

7.3 Ordered pairs in E and what they mean for particle trajectories

A persistent confusion is to treat E as “syntactic derivation.” In the well formalism, E is simpler: it is a set of licensed ordered pairs of states. If $(a,b) \in E$, the system licenses $a \to b$ as an internal transition inside the confined subsystem. This does not by itself assert a surface syntactic rule; it asserts admissibility of state-to-state motion given the current restriction and measurement protocol.

For particle material, typical E edges include: free particle → enclitic (binding), enclitic → bound exponent (stabilization), particle → particle (alternation), and particle → elimination (loss). For bound suffix atoms, typical E edges include: allomorph A → allomorph B (competition), suffix variant → captured (absorption), and suffix+host → fused attractor (fusion).

In §8 we provide a concrete checklist for extracting E edges from corpus distributions or judgment paradigms, staying within the manuscript’s measurement constraints.

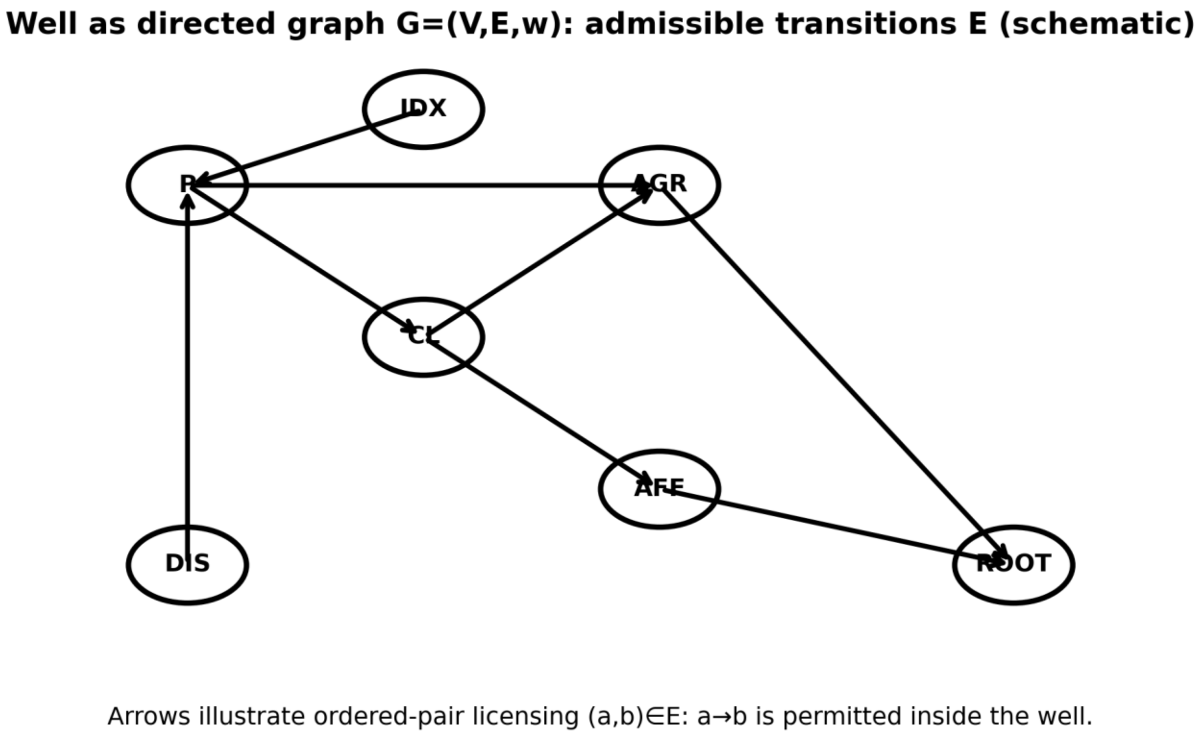
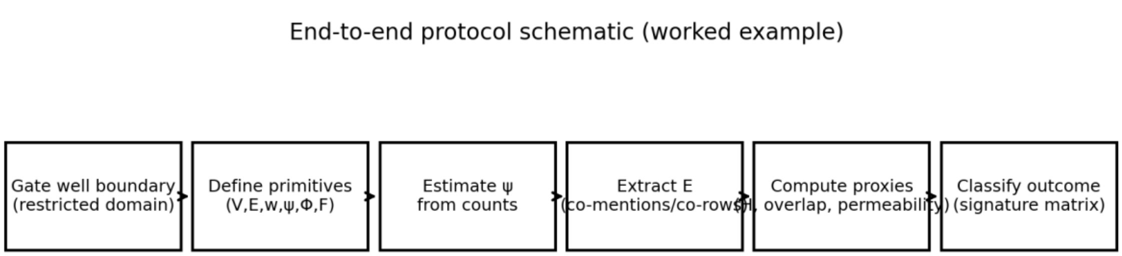


Figure 5 — Directed graph interpretation of the well: E is a set of licensed ordered pairs (a,b) enabling a→b transitions (schematic).

7.4 End-to-end worked example (miniature): the atom "-s"

This worked example runs the protocol end-to-end on a deliberately small, document-internal dataset. It is a methodological demonstration (how to compute ψ, E, and signature proxies) rather than a claim about corpus frequencies. Replace the proxy counts with real corpus/attestation counts in the measurement program.

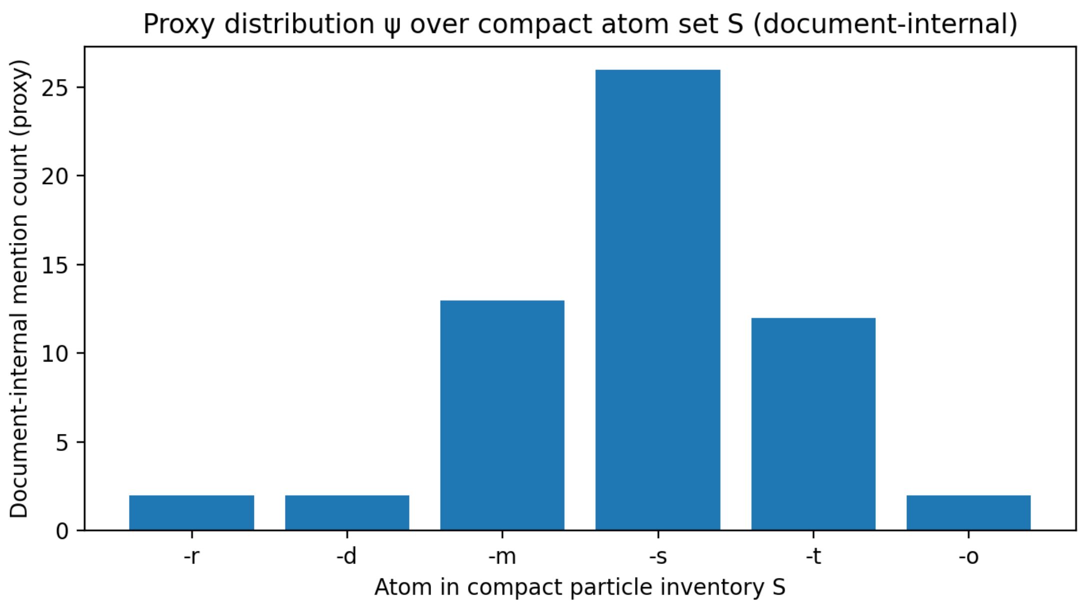
Figure 6A — End-to-end protocol schematic used in the worked example (schematic).



Gating condition (well boundary). Restrict the well to the compact particle atom set S used in this manuscript's PIE analysis (e.g., {-r,-d,-m,-s,-t,-o}) and the environments enumerated by the RelExp_L licensing inventory (above). Within this boundary, V is the atom inventory and E tracks admissible alternation/competition relations among atoms.

Table 6A — Proxy ψ estimation over compact atom set S (document-internal mention counts).

Figure 7A — Proxy distribution ψ over S (document-internal counts; schematic proxy).



Proxy metrics. Entropy H(ψ) = 1.98 bits (effective states ≈ 3.96). For the atom -s: coverage (permeability proxy) = 0.77 of all loci, and its largest overlap competitors are -t (J=0.45) and -m (J=0.45). In a diachronic dataset, increasing coverage and increasing entropy would diagnose Diffusion; a sustained dominance of one atom with asymmetric competitor loss would diagnose Absorption; peak-count reduction in the atom-condition distribution would diagnose Fusion.

Table 6B — Minimal quantitative add-on: |S|, recurrence, and a stability proxy (paragraph vs table contexts).

Outcome labeling (signature matrix). In this miniature snapshot, -s exhibits (i) very broad contextual reach (high permeability proxy) together with (ii) strong dominance in ψ. This combination matches a Diffusion→Absorption pathway: the atom spreads across environments while also capturing mass from close competitors (-t/-m) rather than remaining a low-mass optional particle.

Connection to the paper's predictions. The worked example illustrates how P3 (particle-origin bound morphology manifests as a depth-trajectory inside Particle_L wells, not as a clausal primitive) is operationalized: we measure spread and capture signatures using ψ, E-related overlap, and permeability/entropy proxies while maintaining the Non-collapse axiom (TYPE □ DEPTH).

7.5 Reproducible micro-dataset (RDS-1): Appendix-D deictic particle exemplars

This section adds a fully reproducible measurement layer drawn only from the source material. RDS-1 is extracted from Table 11 (Appendix D) in the MORPHIX, which lists cross-branch exemplar forms for deictic/relational particles. Each exemplar is treated as a token in a restricted domain D, indexed by branch. The dataset is distributed as a CSV alongside this paper (RDS1_AppendixD_DeicticExemplars.csv) so the counts and derived indices below can be regenerated exactly.

Extraction rule (deterministic). For each branch row in Table 11, split exemplars by comma/semicolon separators; clean trailing punctuation; record (branch, form). For the minimal atom-family analysis below, assign a family label by the initial segment in the surface string (T = {t,d,þ,ð… dental onset}, K = {k,c,q,g… velar onset}, and additional bins H/HW/W/A/U/OTHER by initial symbol class). This family assignment is an analysis layer; it does not retype the items as roots. It is only a V-partition used to compute ψ and diagnostics.

Table 7.5A. RDS-1 counts: exemplar tokens by branch and initial-family bin.

Table 7.5B. RDS-1 derived diagnostics by atom-family (ψ over branches).

Figure 7.5A. Entropy proxy (H) for ψ across branches, by atom-family.

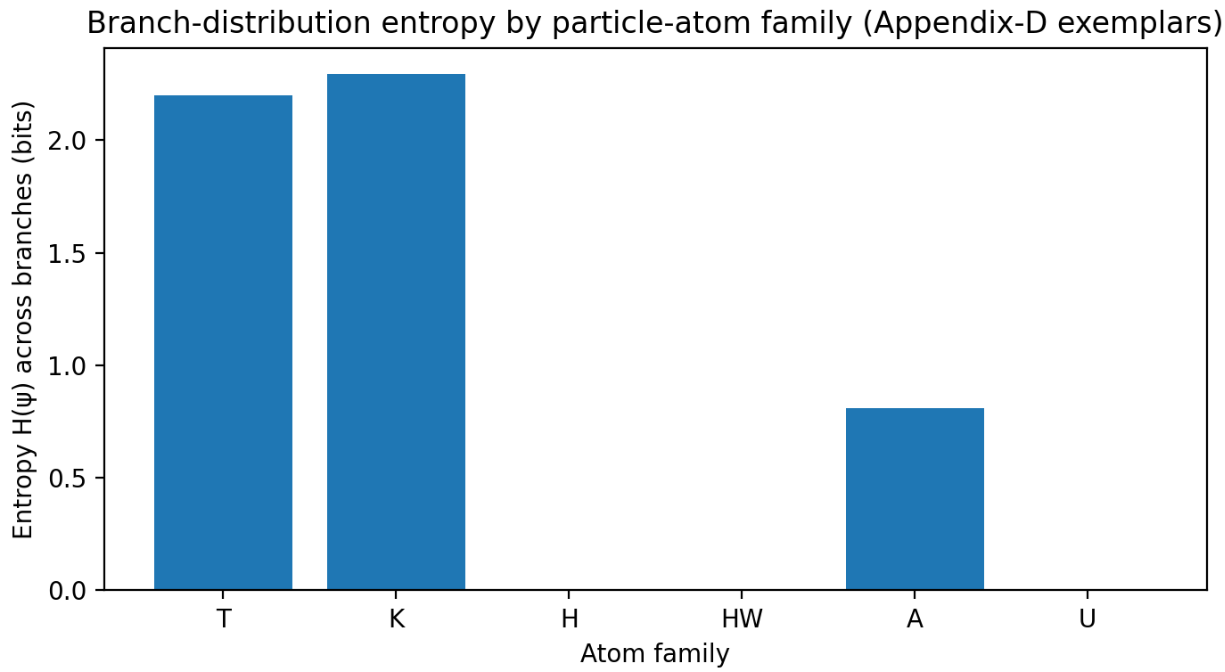
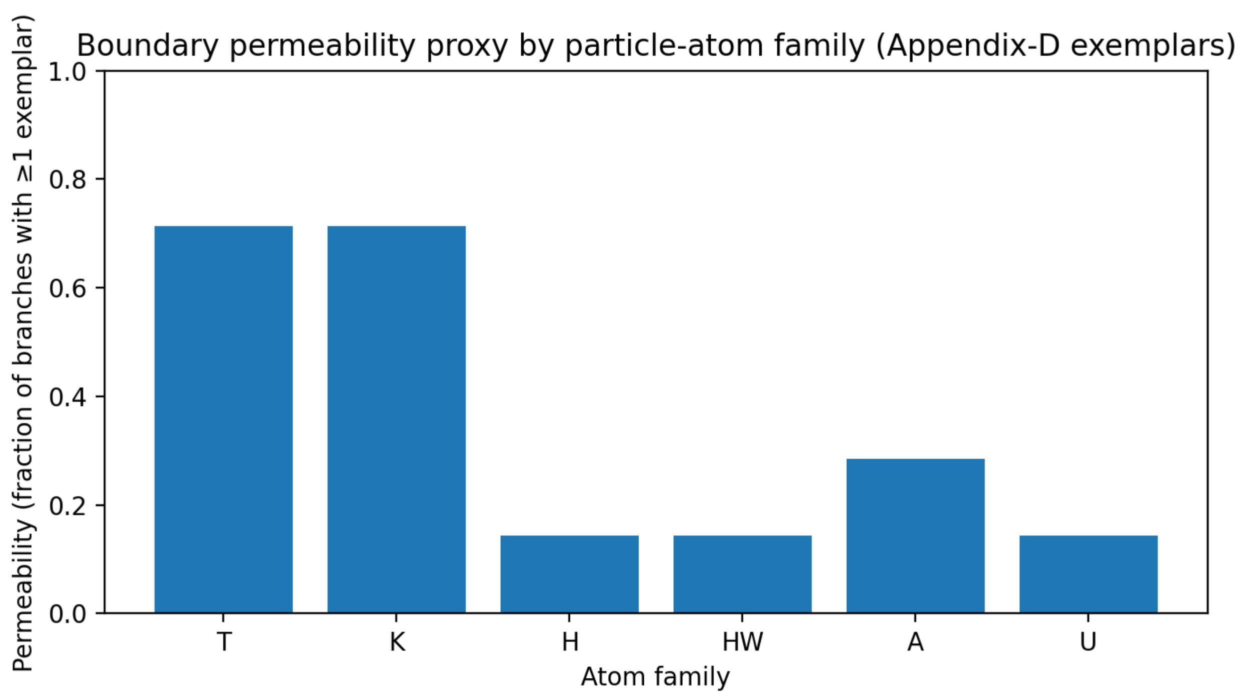


Figure 7.5B. Boundary permeability proxy: fraction of branches with ≥1 exemplar, by atom-family.



Outcome classification on real counts (signature matrix application). Treat each family as a peak over the branch axis, with ψ_fam(b) proportional to the count of exemplars in branch b. Families with high permeability and higher entropy instantiate the Diffusion signature (broadening/leakage across boundaries). Families with low entropy and low coverage instantiate Absorption-like confinement (mass concentrated in one basin/branch subset), and absence corresponds to Elimination. In RDS-1, T and K are diffusion-like (broad coverage), while H/HW are absorption-like (high concentration in a single branch), exactly matching the functional-inventory behavior claimed for particle-type primitives under cross-branch persistence.

Reproduction capsule. The CSV shipped with this paper contains the token list. Recompute Tables 7.5A–B by grouping and normalizing counts, then compute entropy $H(\psi)=-\sum_b \psi(b)\cdot\log_2 \psi(b)$ and permeability as the fraction of branches with nonzero mass.

family	total_exemplars	branches_covered	permeability	entropy_bits
T	13	5	0.714	2.200
K	13	5	0.714	2.295
H	3	1	0.143	-0.000
HW	3	1	0.143	-0.000
A	4	2	0.286	0.811
U	4	1	0.143	-0.000

branch	A	GREEK	H	HW	K	OTHER	T	U
Anatolian	1	0	0	0	0	1	1	0
Celtic	0	0	0	0	3	2	0	0
Germanic	0	0	3	3	0	0	4	0
Greek	0	11	0	0	0	0	0	0
Latin	0	0	0	0	2	0	0	4

Sanskrit	3	0	0	0	3	3	3	0
Slavic	0	0	0	0	2	2	3	0
Tocharian	0	0	0	0	3	0	2	0

Atom a	**Count c(a)**	**ψ(a)=c/Σc**	**Loci |L(a)|**	**Coverage |L(a)|/|L|**	**Jaccard with -s**
-r	2	0.035	2	0.077	0.048
-d	2	0.035	2	0.077	0.048
-m	13	0.228	12	0.462	0.455
-s	26	0.456	20	0.769	1.000
-t	12	0.211	12	0.462	0.455
-o	2	0.035	2	0.077	0.048

Atom set size |S|	**Atom a**	**c_par(a)**	**c_tab(a)**	**ψ_par(a), ψ_tab(a)**	**Stability proxy 1-|Δψ|**
6	-r	2	2	0.032, 0.143	0.889
	-d	2	2	0.032, 0.143	0.889
	-m	14	3	0.222, 0.214	0.992
	-s	29	3	0.460, 0.214	0.754
	-t	14	2	0.222, 0.143	0.921
	-o	2	2	0.032, 0.143	0.889

8. Inferences, falsifiers, and measurement templates

The manuscripts emphasize that the well formalism is not a metaphor but a measurement program: ψ is estimated from counts or judgments; E is estimated from attested transitions (including gaps); and dynamic outcomes are diagnosed by changes in entropy, overlap, permeability, and curvature over time or across conditions. This section summarizes what the particle-origin thesis predicts, what would falsify it, and how to operationalize the relevant indices without leaving the ULD/UNS logic.

Throughout, “evidence” is defined as a mismatch between particle-type predictions and root-type predictions when the same exponent is tracked across depth regimes.

8.1 Diagnostic indices (operational definitions)

Entropy. For a distribution ψ over a finite inventory V, define $H(\psi)=-\sum_{v\in V} \psi(v) \log \psi(v)$. Diffusion predicts H increases as mass spreads across more states; elimination or capture can decrease H after mass relocates into fewer peaks.

Overlap. For two distributions ψ_1 , ψ_2 over the same V (e.g., two periods, two registers, two conditioning environments), define overlap $O(\psi_1, \psi_2)=\sum_{v\in V} \min(\psi_1(v),\psi_2(v))$. Fusion is expected to show a transient increase in overlap among previously separated peaks, followed by stabilization.

Boundary permeability. For an edge set E and a partition of V into “inside” and “outside” (or into basins), permeability can be approximated by the mass of transitions that cross barriers relative to those that remain within basins. Operationally, this requires an E extraction protocol: identify minimal alternation edges and count co-occurrence transitions. Diffusion predicts increased cross-barrier mass; deep stabilization predicts reduced cross-barrier mass (higher effective w).

Curvature. The manuscript uses curvature informally as a summary of how peaked distributions are (sharp basins vs flat plateaus). A simple operational proxy is the second-difference curvature of the sorted ψ values or the kurtosis of ψ. Collapse predicts a discontinuity: a sudden change in curvature accompanied by regime-threshold behavior.

8.2 Outcome-by-index signature matrix

Outcome	Entropy H	Overlap O	Permeability P	Curvature / peakiness
Drift	~ (moderate)	~	~	centroid shifts (peak moves)
Diffusion	↑	~ or ↓ (depends)	↑	↓ (flatter)
Fusion	↓ after merge	↑ then stabilizes	↓ (post-merge)	↑ (single sharper basin)

Absorption	↓	↓ for losing peak	↓	↑ for winning basin
Elimination	↓ or ~	~	~	↓ variance after removal
Collapse	spike / discontinuity	discontinuity	spike / discontinuity	discontinuity

8.3 Particle-origin predictions for Indo-European bound atoms

Prediction P1 (Closed-class constraint). The compact suffix atoms behave as a small inventory: $|S|$ is small, and new members enter S rarely. This yields strong ψ peaks and strong cross-construction recurrence.

Prediction P2 (Relational licensing). The compact atoms are licensed in relational environments (deixis, indexicality, discourse, clausal modification) rather than as lexical heads. This aligns with RelExp_L and manifests as high conditional probability of occurrence in relational slots.

Prediction P3 (Stabilization pathway). When an atom becomes bound, the time profile should show diffusion → absorption/fusion: entropy rises during spread, then peak count decreases as the bound exponent stabilizes into a deeper attractor.

Prediction P4 (Restricted alternation). If the particle layer is non-ablauting or restricted in alternation, the allowed edge set E for particle atoms should be sparse compared to root alternation graphs.

Prediction P5 (Non-primitive clausal behavior). Apparent clause-operator effects should correlate with Φ coupling patterns (co-variation with clause environments) rather than with a dedicated clause-primitive status; perturbing Φ (e.g., changing conditioning environments) should change the effect.

8.4 Falsifiers (what would refute the particle-origin claim)

F1. A compact suffix atom behaves as a root: it is open-class, head-capable, supports derivational expansion as a head, and participates freely in compounding. This would satisfy Root_L and contradict Particle_L.

F2. A compact atom's distribution is insensitive to relational environments (fails RelExp_L) and is instead anchored as a lexical nucleus. This would contradict the relational licensing assumption.

F3. Diachronic stabilization of the atom does not show the expected wave-dynamic signatures (no diffusion during spread; no absorption/fusion during binding; no curvature changes). This would undermine the morphodynamic pathway explanation and would require a different internal mechanism.

8.5 Minimal measurement protocol templates

Template T1 (ψ estimation). Choose a restricted domain (the well's gating condition), list the inventory V of candidate states, and estimate ψ as normalized token counts or normalized judgment frequencies.

Template T2 (E extraction). Define admissible directed transitions as ordered pairs (a,b) that are attested as alternation, competition, or co-occurrence transitions under the manuscript's definition. Record gaps explicitly: $(a,b)\notin E$ where transitions are unattested despite comparable mass elsewhere.

Template T3 (Outcome diagnosis). For two time slices or two conditioning environments, compute H, O, and a permeability proxy, and compare to the signature matrix. Classify the change as drift, diffusion, fusion, absorption, elimination, or collapse.

Template T4 (Particle predicate audit). For each candidate atom a, explicitly check RelExp_L(a) and Root_L(a) conditions with language-internal diagnostics. Only then project to Particle_L(a) and interpret the wave outcomes.

9. Conclusion

The internal claim developed here is that a single confinement geometry can host both free discourse modifiers and bound suffixal atoms without treating the latter as clause primitives. The difference between “free particle,” “clitic-like particle,” and “bound exponent” is modeled as a depth regime inside the same well, not as a change in primitive ontological type.

The proof-oriented contribution is to separate three components: (i) a particle predicate that classifies primitive type (RelExp $\wedge$ ¬Root), (ii) a confinement geometry (V,E,w,ψ) that models positional stabilization, and (iii) wave-dynamic diagnostics that distinguish outcomes (drift/diffusion/fusion/absorption/elimination/collapse). Under this separation, the particle-origin interpretation of -s (and its extension to the compact suffix set) is not a metaphor but a falsifiable hypothesis: the pathway from free relational material to stabilized bound morphology should have measurable signatures in ψ-movement and E licensing.

Finally, the MORPHIX program provides a concrete closed-class inventory and a feature-geometry scaffold that make the particle layer explicit. That scaffold is the right place to locate pronoun stems and discourse modifiers; it prevents misplacement of pronouns as deep root-anchored entities and keeps the ontology clean: only root-anchored referential nuclei occupy the deepest stable basins, while particle material can stabilize as bound atoms without becoming a root.

Appendix A — Well formalism in compact notation

A.1 Well as a weighted directed graph. A well is a triple (V,E,w) where V is a finite set of states, $E \subseteq V\times V$ is a set of admissible directed edges (ordered pairs), and $w:E\rightarrow\mathbb{R}^+$ assigns a weight/cost to each edge. The potential V(x) can be treated as a continuous abstraction of these discrete weights along a depth coordinate x, but the discrete form suffices for measurement.

A.2 Distribution object. ψ is a probability distribution over V: ψ:V→[0,1] with ∑_{v∈V} ψ(v)=1. In practice, ψ is estimated from token counts or normalized judgments under a gating condition.

A.3 Dynamics. A trajectory τ is a typed sequence of edges $(v_0, v_1), (v_1, v_2), \ldots$ in E. Wave outcomes classify families of such τ under changes of ψ and under changes of the effective permeability induced by w and Φ coupling.

A.4 Coupling field. Φ encodes how contexts couple states; operationally it is a conditional distribution over V given context features. Changes in Φ shift ψ mass and can alter which τ are licensed or preferred.

Appendix B — Worked classification template for a compact atom (illustrative)

The purpose of this appendix is procedural, not empirical. It shows how one would classify an exponent a (e.g., -s) without assuming it is a clause primitive.

Step 1: Choose restriction. Define the restricted domain for the well (e.g., pronominal/adverbial morphology in a curated corpus of Indo-European languages).

Step 2: Define V. Inventory candidate states: stems and atoms, including a.

Step 3: Estimate ψ. Compute normalized counts of a across contexts.

Step 4: Extract E. Record which transitions (a,b) are attested: e.g., free particle ↔ enclitic, enclitic ↔ bound exponent, allomorph alternations, host+atom resegmentations.

Step 5: Compute indices. Compare H, overlap, and permeability across conditions/time slices.

Step 6: Diagnose outcome. Use the signature matrix to classify drift/diffusion/fusion/absorption/elimination/collapse.

Step 7: Apply particle predicate audit. Check RelExp_L(a) and ¬Root_L(a). If Particle_L(a) holds, interpret the outcome as particle-origin stabilization rather than as clause-primitive behavior.

Appendix C — MORPHIX inventory and criteria (summary extraction)

This appendix summarizes the MORPHIX criteria and inventory claims as they are used in the present article. MORPHIX proposes that pronoun stems and deictic/adverbial stems in PIE reflect an older particle layer: closed-class, non-derivable, non-compounding, and with restricted alternation behavior.

For convenience, the compact suffix set (e.g., -r, -d, -m, -s, -t, -o) is treated as the bound-morphology face of the same particle inventory.

Appendix D — Diagram index

Figure 1 — Well cone and three zones (surface closure; augment channel; root nucleus).

Figure 2 — Particle→clitic→suffix stabilization pathway.

Figure 3 — Outcome signatures as time profiles (schematic).

Figure 4 — MORPHIX feature-geometry scaffold (schematic).

Figure 5 — Directed graph interpretation of E as ordered pairs.

Appendix E — Particle vs Root diagnostic matrix (ontology audit)

This appendix provides a compact audit matrix for the two predicates used throughout: Root_L and RelExp_L. It is designed to be filled with language-internal evidence (corpus distributions, judgment paradigms, or descriptive diagnostics) without importing external theoretical categories.

Diagnostic	Root_L expectation	RelExp_L expectation	Operational handle
Open-class expansion	Yes (new members common)	No (closed class)	inventory size over time; neologism intake
Head-capable derivation	Yes (hosts affixes as head)	No (attaches/conditions; not head)	derivational family structure
Compounding	Permits productive compounding	Resists compounding	attested compounds; productivity tests

Distributional anchoring	Anchored to lexical nuclei	Licensed by relational contexts	conditional distribution by context
Alternation graph density	Dense alternation network	Sparse restricted alternations	edge count $\|E\|$; alternation inventory
Cross-category portability	Limited portability	High portability (clausal/discourse)	occurrence across environments

Appendix F — Depth, permeability, and stabilization (formal lemmas)

Lemma F1 (Permeability monotonicity under increasing weight). Let P be a permeability proxy defined as the sum of transition mass across barrier edges divided by total transition mass. If w on barrier edges increases while ψ is held fixed, then P decreases monotonically under any stochastic transition model whose probability of traversing an edge e is non-increasing in w(e).

Proof sketch. For a transition model with probability proportional to exp(−w(e)), increasing w(e) reduces its contribution. The ratio defining P therefore decreases as barrier edges are downweighted. This lemma is the formal reading of "deep stabilization reduces boundary permeability."

Lemma F2 (Absorption as asymmetric basin capture). In a two-peak system where ψ mass is partitioned into peaks A and B, if w favors transitions from B to A more than from A to B, then the stationary distribution concentrates on A. This is the formal caricature of absorption: a deeper basin captures a shallower basin.

Corollary. When a particle element becomes bound as a suffixal atom, the stabilization corresponds to an increase in effective barrier weights and/or an asymmetric Φ coupling that biases transitions into the host basin. Hence absorption/fusion are the expected outcomes for particle→bound pathways.

Appendix G — Eigenstates and strata (spectral reading)

The manuscript uses "eigenstates" to refer to stratified stable modes: clustered distributions or plateaus that behave as stable layers. Formally, if a transition operator T is defined on ψ (e.g., via the weighted edge dynamics), eigenstates are fixed points or low-variance modes of T. The importance for particle stratification is that particle inventories can form their own eigenmodes (stable surface strata) even when their material later participates in deep stabilization as bound atoms.

In the ULD/UNS reading, eigenmodes do not impose categories; they report stability: a stratum is an eigenstate when the system returns to it under perturbation.

Appendix H — Outcome decision tree

The diagram below is a procedural decision tree for classifying observed changes in ψ into canonical outcomes using the indices defined in §8.1 (schematic).

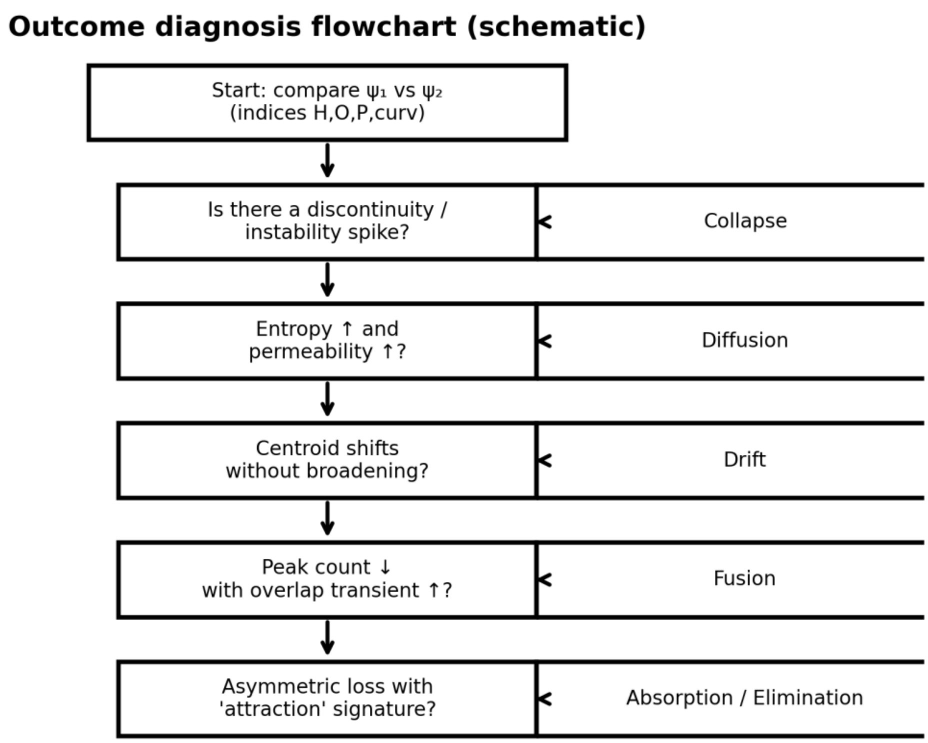


Figure 6 — Outcome diagnosis flowchart based on discontinuity, entropy/permeability, centroid shift, and peak-count change.

Internal references (sources)

Morphodynamics Manuscript; Synopsis Morphodynamics ULD_UNS.

MORPHIX Feature Geometry, Vocabulary Insertion, and the Particle Stratification of PIE Pronominal and Adverbial Morphology.

Appendix I — Micro-evidence layer from MORPHIX (extracted tables)

This appendix imports a minimal quantitative layer already present in the MORPHIX manuscript. It is included to keep the particle-stratification thesis grounded in measurable outputs, even when the present article's figures are schematic. The key point is not the absolute numbers, but the pattern: introducing a compact particle inventory (Morphix stems) improves reconstruction/coverage metrics relative to a root-only baseline, and the compatibility matrix for stems is branch-sensitive in a way consistent with a restricted particle inventory.

I.1 Corpus proxy reconstruction (Table 8; Root+Morphix)

Corpus proxy	N tokens	N modeled	Coverage	Modeled cell accuracy	Unique modeled cells
Old English	52	41	0.788	1.0	32
Vedic/Sanskrit	77	72	0.935	1.0	70

Reading. The Root+Morphix model shows high modeled-cell accuracy in both corpus proxies and relatively high coverage, suggesting that a compact particle stem inventory helps explain token distributions without requiring a large open-class root inventory.

I.2 Baseline reconstruction (Table 8A; Root-only)

Corpus proxy	N tokens	N parsed	Precision	Recall	F1	Cells reconstructed
Old English	52	41	1.0	0.788	0.882	29
Vedic/Sanskrit	77	72	1.0	0.935	0.966	66

Reading. The Root-only baseline achieves high precision but lower recall and lower reconstructed-cell counts, consistent with the claim that a purely root-driven inventory undergenerates the structured particle-stem space.

I.3 Branch × stem compatibility (Table 12) and its geometric interpretation

Branch	h1e-	h1i-	k^wi-	so-	to-	yo-
Anatolian	1	0	0	1	1	0
Celtic	0	0	1	1	1	0
Germanic	0	1	1	0	1	0
Greek	0	0	1	1	1	1
Latin	0	0	1	0	0	1
Sanskrit	0	1	1	0	1	1
Slavic	0	0	1	0	1	1
Tocharian	0	0	1	0	1	0

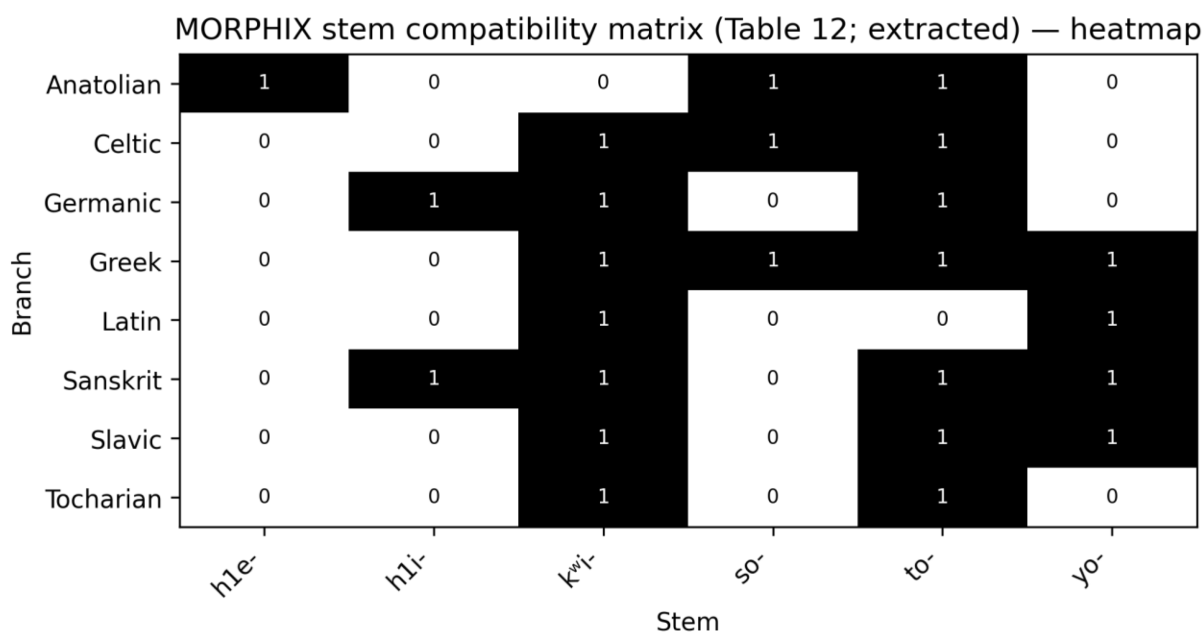


Figure 7 — Heatmap rendering of Table 12 (stem compatibility by branch).

Geometric reading. In the well formalism, a compatibility matrix can be treated as a projection of E: a 1 indicates that a transition path or licensing relation exists between the stem state and the branch-defined subsystem, while a 0 indicates a gap (an unattested or disallowed transition). The branch structure thus corresponds to different subgraphs or differently weighted barriers, which is precisely what a confinement geometry is designed to encode.

Appendix J — Cost and parsimony formalism (how Morphix compresses a particle inventory)

The MORPHIX manuscript motivates its inventory using a compression argument: a compact particle stem set plus compact suffix atoms can reduce descriptive cost by explaining a wide distributional footprint with fewer stipulations. This fits directly into the well parameters: $|V|$ is constrained, E is sparse but structured, and w assigns stability/penalty costs that trade off with data fit.

A generic cost functional can be written as: $C(\text{Model})=C\_\text{structure}(V,E,w)+C\_\text{fit}(\psi \mid \text{data})$. C_structure penalizes large inventories and dense transition graphs; C_fit penalizes mismatch between predicted and observed ψ. Under this reading, a “particle inventory” that is small but portable can lower both terms: it lowers C_structure by keeping V small, and lowers C_fit by capturing cross-construction recurrence.

Lemma (Compression advantage of portable particle atoms). If a particle atom p is used across k environments that would otherwise require k separate environment-specific exponents, then introducing p reduces C_structure by collapsing k vertices into 1, while C_fit can be maintained by tuning Φ coupling to route p into the appropriate environments. This is the formal skeleton of the manuscript’s parsimony claim.

10. Discourse and clausal modifiers as particle regimes (extended discussion)

The particle-origin thesis is easiest to see in discourse and clausal modifiers because the “free vs bound” contrast is transparent.
A free modifier (e.g., a discourse particle, a clause-edge adverbial, a focus particle, an evidential marker) is morphologically autonomous or weakly bound,
yet it can exert systematic effects on clausal interpretation. Traditional accounts often elevate such material to “clausal primitives” on the basis of its scope.
Within Morphodynamics, that inference is not licensed: scope is an effect of coupling (Φ) and transition licensing (E), not a definition of primitive type.

In confinement terms, discourse/clausal modifiers occupy the surface closure region of the well. Their hallmark properties—high optionality, high contextual mobility, and sensitivity to register/discourse conditions—are precisely what the well predicts for surface material: barriers are permeable (effective w is low), the distribution is allowed to broaden (diffusion), and coupling to global contexts is strong (Φ is high-rank). This is the correct regime for "free" particles: they can be inserted, omitted, repeated, stacked, or repositioned without forcing a deep reparameterization of the host nucleus. The same surface regime can produce apparent clause-level systematicity because the clause interacts with the well at the boundary. Operationally, this means that the clause does not "contain" the particle as a skeleton primitive. Rather, the clause reads the particle as a boundary condition: the particle shifts Φ-coupling, thereby redirecting which transitions are preferred inside the well and which attractors are selected as stabilized outputs. This is exactly the sense in which clausal effects are emergent. The emergent effect can be strong (categorical) if Φ coupling is strong, even when the particle remains surface material.

The core typological move is to unify weakly bound modifiers with bound exponents via the same dynamical pathway. A particle may begin as free discourse material (surface), become enclitic under repeated adjacency (middle), and finally stabilize as a bound exponent (deep). At each step the primitive type can remain particle (Particle_L holds), but the confinement changes: ψ mass becomes more concentrated, and barrier weights increase. This produces the familiar historical path "particle → clitic → affix," but crucially: it also predicts that "affix" does not entail "clause primitive."

10.1 Modifier typology mapped to well zones (schematic but operational)

Modifier type	Typical binding	Well zone	Likely dynamics	Diagnostic expectation
Discourse particle / stance marker	free	surface closure	Diffusion ↔ Elimination	high optionality; entropy sensitive to register
Clause-edge adverbial / connective	weakly bound	surface→middle channel	Drift + Diffusion	centroid shifts by context; permeability high
Focus particle / contrast marker	weakly bound	middle channel (Φ strong)	Absorption risk	competition/capture among near-synonyms;

				asymmetric loss
Evidential / modal enclitic	clitic-like	middle channel	Absorption / Fusion	loss of free variant; bound stabilization with overlap peak
Bound clause-marking exponent (historical particle)	bound	deep	Fusion / Absorption	peak-count reduction; permeability decreases; increased stability

The typology is "schematic" in that it does not assert specific language facts, but it is operational: each row corresponds to a predicted combination of indices (H, O, P, curvature) and to a predicted location of ψ mass in the well.

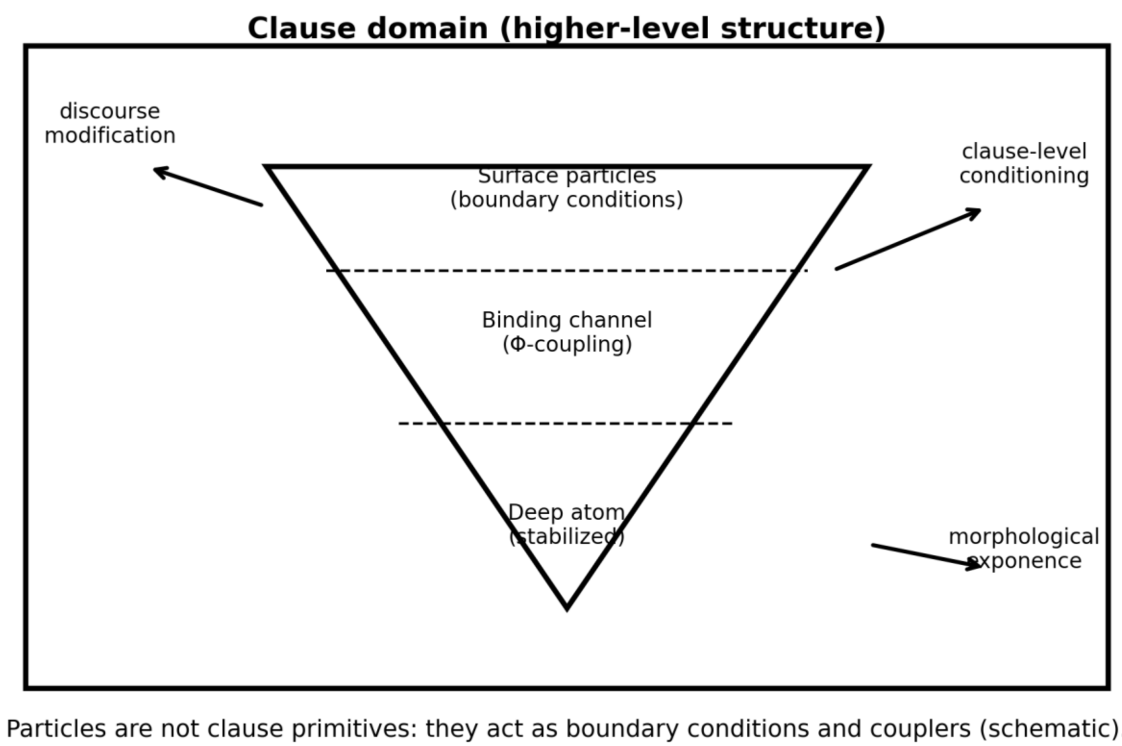


Figure 8 — Clause embedding of the well: particles act as boundary conditions and couplers rather than clause primitives (schematic).

11. Indo-European “primitive” suffix atoms as particle-origin elements (case synthesis)

The Indo-European examples targeted in this article (-s, -t, -m̥) are “primitive” in a very specific sense: they are short, frequent, and appear widely across pronominal/adverbial and related paradigms. Because they participate in structures that correlate with clausal interpretation (relative formation, indexical reference, deixis, clause linkage), it is tempting to treat them as clausal primitives. The proof sheet rejects this inference and instead classifies -s as particle origin by explicit predicate audit: it behaves as a relational exponent and fails root status.
In that ontology, being suffixal does not change the primitive type. It only changes depth.

The particle-origin analysis is strengthened when the suffix atoms are placed inside a compact inventory rather than treated one-by-one. MORPHIX explicitly motivates a compact set of stems and suffix atoms that recur across pronominal and adverbial systems. The compression idea is simple: if a small number of particle atoms can be combined (via licensed transitions E and coupling Φ) to generate a wide typological footprint, then the particle inventory is both descriptively economical and evolutionarily plausible. In well terms: the system uses a small V with structured E, and it adjusts w and Φ to route mass ψ through the graph.

The key morphodynamic claim is that the “clausal reach” of these atoms is explained by their position at the boundary between particle and clause domains. Particles couple strongly to discourse and to clause environments (Φ has a global scope), so even when the particle itself is not a clause primitive, it can redirect which attractor states are selected in the clause’s local configuration. Historically, repeated coupling plus adjacency can force deeper confinement: a particle becomes enclitic and eventually fuses with the host, yielding a bound exponent that looks “morphological.” But this is a diachronic outcome of confinement, not a primitive reclassification.

A practical advantage of the particle-origin stance is that it turns a philosophical dispute (“is this syntactic?”) into an empirical workflow. One does not ask whether -s is “syntactic.” One asks: does it satisfy RelExp_L and fail Root_L? If yes, it is particle origin. s
Then one asks: does ψ place it in a shallow or deep basin in a given subsystem? That answers whether it behaves free, clitic-like, or bound.

Finally one asks: when the system changes, which wave-dynamic outcome signature is observed (drift/diffusion/fusion/absorption/elimination/collapse)? This yields falsifiable diagnostics rather than terminology. The proof sheet also notes that the same logic applies to -t and -m̥.
Within the present synthesis, this is captured by Theorem 2: for any compact atom a, if it is relational and not root, then it is particle. The empirical burden is to document the relational licensing and non-root status for each atom in the relevant system. MORPHIX contributes exactly such documentation at the level of system architecture: it treats the relevant material as derived from a particle layer, not from lexical roots.

11.1 A diachronic pathway statement in well terms

To make the pathway explicit, suppose p is a discourse particle with Particle_L(p).
At time t_0 , ψ concentrates p at the surface: the particle is optional and mobile. This corresponds to a low effective barrier weight w across the boundary, and high permeability. Under diffusion, ψ(p) spreads to new contexts; under drift, preferences shift toward adjacency with particular hosts.
At time t_1 , repeated adjacency plus strong Φ coupling biases transitions into a clitic-like realization: p is still particle-origin, but it occupies the middle binding channel.
At time t_2 , absorption and fusion occur: the p-peak is captured by a deeper basin associated with a host; peak count decreases and curvature increases; permeability decreases.
The observable result is a bound exponent that is historically particle-origin and dynamically stabilized. This statement is not a claim about any specific language. It is the morphodynamic skeleton that the IE suffix atoms are claimed to instantiate. Empirical work then consists in (i) locating which phase a given atom currently occupies in a given sub-system, and (ii) verifying the predicted signatures.

Appendix K — Proof-sheet formalism for particle origin (summary)

The proof sheet provides a second, more geometric phrasing of particle origin that aligns cleanly with the well formalism. It distinguishes two flows (streams) inside the same morphodynamic space:

(i) a root-stream that moves toward a nucleus of stable entity/event condensates (deep basins), and

(ii) a particle-stream that flows across the surface as relational operators (surface closure material).

In this reading, the minimal "primitive" atoms (-s, -t, -m̥, etc.) are not located in the nucleus as lexical roots; they belong to the particle-stream by default, even if they appear later as bound morphology after stabilization. In other words: "primitive" here means "atomized particle material," not "clause skeleton."

A key methodological axiom in the proof sheet is an exhaustivity constraint: any exponent must be typed as either root-stream material or particle-stream material, and mixed typing is not allowed without an explicit derivation. This is the proof-sheet counterpart of the Particle_L predicate audit used in §3 and §6.

K.1 Two-stream exhaustivity and its consequences

> Axiom (Two-stream exhaustivity; proof-sheet phrasing). Every exponent x belongs to exactly one of two streams: Root_Stream(x) $\oplus$Particle_Stream(x).

Consequence 1. If an exponent is assigned to Particle_Stream, it cannot be treated as a lexical root without an explicit reclassification derivation. This blocks the common slide from "tiny and widespread" to "rootlike" or "clause-primitive."

Consequence 2. Boundness is orthogonal to stream membership. An exponent may be particle-stream and still become bound under confinement. This is equivalent to saying that Particle_L(x) is compatible with deep stabilization.

K.2 ULW cone reading (surface operators vs nucleus condensates)

The proof sheet also presents the ULW geometry as a cone: a nucleus hosting entity/event condensates, orbital layers hosting derived categories, and a surface hosting relational operators (particles). On this picture, the particle inventory is expected to be portable across contexts, because it lives on the surface and couples broadly.

Stream / region	Primitive type	Well zone alignment	Expected behaviors
Nucleus / root-stream	Root_L material (lexical nuclei)	Deep basins	open class; head-capable; stable

			anchoring; rich derivation
Surface / particle-stream	Particle_L material (relational operators)	Surface closure + binding channel	closed class; portability; high Φ coupling; can stabilize into bound atoms
Transition channel	Particle→bound trajectories	Middle band	cliticization; absorption; diffusion prior to capture
Deep stabilization of particle atoms	Bound particle-origin exponents	Deep basins (captured)	fusion; decreased permeability; high recurrence as suffix atoms

K.3 Proof-sheet alignment with Theorem 1

Theorem 1 in the main text is the predicate form of the same typing: if -s behaves as a relational exponent and is not a root, it is particle. The proof sheet's stream geometry provides an independent conceptual justification: particles live on the surface, and the primitive atoms are surface material by default, even when later stabilized as bound morphology.

12. Feature geometry and vocabulary insertion as well-internal operators (extended formalism)

MORPHIX does not merely assert that pronoun stems are "particles." It provides a concrete representational scaffold: a feature geometry that partitions a compact inventory, and an insertion logic that maps feature bundles to surface exponents.

This is directly compatible with the well formalism when feature nodes are treated as latent state coordinates and insertion rules are treated as admissible transitions (edges) that route ψ mass into particular outputs.

12.1 Feature geometry as state-space factorization.

Let V be the inventory of possible exponent realizations in a restricted domain (e.g., pronominal/adverbial stems in PIE reconstructions). A feature geometry provides a factorization of V into a small number of latent factors (nodes), such that each exponent corresponds to a conjunction of feature-node activations. In probabilistic terms, it provides a structured prior over ψ: it prefers distributions that can be generated by a low-rank latent structure rather than by independent unstructured listing. In confinement terms, the feature geometry controls which edges are even candidates for inclusion in E. If two exponents differ only by one feature node, a transition edge between them is licensed as a minimal alternation; if they differ by many independent nodes, a direct transition is disfavored (high w) or disallowed (edge absent).

This yields a direct prediction: particle inventories should have sparse and structured alternation graphs because their feature geometry is shallow and atomized.

12.2 Vocabulary insertion as τ-licensing.

Insertion rules can be read as a typed mapping $\iota: F \to V$ from feature bundles to exponent states. A particular utterance context sets a target feature bundle $f \in F$; insertion chooses the exponent $v = \iota(f)$.

Within the well, this corresponds to a trajectory τ that begins at a context-conditioned state and ends at an output exponent state. If multiple exponents compete for the same f, their competition is modeled as drift or absorption depending on whether the preference shift is gradual or capture-like.

12.3 Why this matters for the particle-origin thesis.

The worry about "category inflation" arises when particle material is treated as syntactic primitives because it correlates with clause-level effects. The feature-geometry / insertion reading blocks that inflation: the particle inventory is already representationally rich in a well-internal way. It can condition clause-level interpretation via Φ coupling without being reclassified as clause skeleton. This is the correct division of labor: clause structure is not rebuilt each time a particle appears; instead, the particle changes which well-internal features are activated and which transitions are preferred.

12.4 A testable consequence: alternation locality.

If particle inventories are feature-geometry structured, then alternations among particle atoms should be local in feature space. Operationally: edges in E should overwhelmingly connect states that differ by one or a very small number of feature-node changes. This can be tested with the compatibility matrices and reconstruction tables already present in MORPHIX: the same small stem set can be routed into different branches by toggling a small number of features, rather than by introducing new roots.

12.1 Locality-of-alternation matrix (template)

To operationalize alternation locality, define a feature distance d_F(v_i,v_j) as the number of feature-node differences between two exponents. Then compute the empirical edge set E from alternation/co-occurrence and evaluate the distribution of d_F across edges. The particle-inventory prediction is that d_F is concentrated near 1; the root-inventory prediction is that d_F is broader.

Quantity	Definition	Particle-inventory expectation	Root-inventory expectation
d_F(v_i,v_j)	feature-node Hamming distance	small on most edges	no strong bias
Pr(d_F=1 | (v_i,v_j)∈E)	edge-locality probability	high	moderate/low
Mean d_F over E	average locality	low	higher

Appendix L — Pseudocode for extracting ψ, E, and indices (operational workflow)

The following pseudocode is meant as a direct operationalization of §8.5. It is written in a neutral form so it can be implemented in any environment.

Inputs:

• Data: tokens with context features (time slice, register, construction, branch, etc.)

• Candidate inventory V: list of exponent states to track (stems, particles, suffix atoms)

• Gating condition: restriction that defines the well (domain boundary)

Workflow:

1. Restrict data by gating condition to obtain well-internal token stream.
2. Estimate ψ by counting tokens per $v \in V$ and normalizing.
3. Extract E by identifying admissible ordered pairs:
   - alternation edges: v_i competes with v_j in matched environments,
   - co-occurrence edges: v_i appears adjacent to v_j in consistent configurations,
   - resegmentation edges: v_i+host maps to v_j (fused output).

   Record both attested edges and systematic gaps.
4. Compute indices:

   $H(\psi)=-\sum \psi(v) \log \psi(v)$;

   $O(\psi_1 ,\psi_2)=\sum \min(\psi_1 (v),\psi_2 (v))$;

   Permeability proxy P from cross-barrier transitions;

   Curvature proxy from sorted ψ peakiness.
5. Diagnose outcome using the signature matrix and the decision tree.
6. Audit particle predicate:

   check RelExp_L(v) and Root_L(v) using diagnostics (Appendix E);

   assign Particle_L(v) where appropriate.
7. Interpret: decide whether the observed change is particle diffusion/stabilization or root-driven restructuring.

Outputs:

• ψ distributions per condition/time slice

• edge list E with weights
• classified outcome labels with supporting indices
• particle vs root typing report

Appendix M — Object-language vs metalanguage control (avoiding category inflation)

The manuscripts consistently separate the object-language behavior of exponents from the metalanguage used to describe them. This control is critical in the particle-origin thesis because a common reviewer objection is that "if it has clause scope, it must be syntactic."
That objection collapses object-language scope behavior into a metalanguage primitive categorization. Within the well framework, scope-like behavior is modeled as coupling and transition selection. A discourse particle can alter which attractor is selected downstream without being a primitive clause operator. To keep the analysis disciplined, each claim should be stated twice: (OBJ) as a statement about distributions (ψ), admissible transitions (E), and indices; and (META) as a statement about the ontological typing predicates (Root_L, RelExp_L, Particle_L).

Recommended reporting format:
• OBJ claim: "In environment C, ψ mass shifts from v_i to v_j; entropy rises; permeability rises; edges (v_i,v_j) become active."
• META claim: "v_i satisfies RelExp_L and fails Root_L, therefore it is particle-origin; the change is diffusion prior to stabilization."

This separation ensures that the analysis does not overgenerate primitives. It also makes falsifiers transparent: an OBJ change can be real while the META typing can be wrong, and vice versa. The particle-origin thesis is thus evaluable without terminological drift.

Appendix N — Formal coupling calculus for particles (math-oriented extension)

This appendix gives a minimal formal calculus that makes the "emergent clause effects" statement precise inside the well formalism.

The aim is to show that clause-level differences can be induced by particle material through Φ-coupling without postulating particle-as-clause-primitive.

N.1 Context-conditioned distributions.

Let C be a finite set of clause environments (contexts), and let V be the well state inventory.

Define Φ as a family of context-conditioned distributions over V:

$\Phi\_c(v) = \Pr(v \mid c) \quad \text{for } c\in C,\ v\in V,$

with $\sum\_\{v\in V\}\ \Phi\_c(v)=1$ for each c.

Φ is empirically estimated by conditioning token counts or judgments on the context label c.

N.2 Particle intervention as boundary conditioning.

Let p be a particle element (Particle_L(p)).

Model the particle's presence in a clause as a boundary-conditioning operator B_p acting on Φ:

$(B\_p\ \Phi)\_c(v) = \Phi\_c(v \mid p)$.

Operationally, this is just conditioning on the presence of p.

Importantly, B_p does not change the clause's primitive skeleton; it changes which well states are activated and with what probabilities.

N.3 Output selection as an attractor map.

Let A:Δ(V)→V be an attractor selection functional that maps a distribution to its dominant basin (or to a ranked list of basins).

For example, A could return argmax_v ψ(v), or it could return the top-k peaks after smoothing by w.

Then the observable exponent output under context c is:

$\text{out}(c) = A(\Phi\_c)$.

With the particle present:

$\text{out\_p}(c) = A((B\_p\ \Phi)\_c)$.

If out_p(c) ≠ out(c), the particle induces a systematic difference in outputs.

This difference is "clause-level" in the sense that it correlates with clause environment c, but it is mediated entirely by well-internal conditioning.

N.4 Emergent scope without clause primitives.

To connect to scope-like behavior, let I be an interpretation functional that maps outputs to an interpretation space Σ.

Then:

$[\![\text{clause}]\!](c) = I(\text{out}(c))$,

$[\![\text{clause}]\!]\_p(c) = I(\text{out\_p}(c))$.

The difference $\Delta(c)=[\![\text{clause}]\!]\_p(c)-[\![\text{clause}]\!](c)$ can be nontrivial even if p is not a clause primitive, because p changes Φ and therefore changes which attractor is selected.

This is the formal statement of “emergent clause effects via coupling.”

N.5 Linking to wave dynamics.

Over time or across regimes, Φ and ψ may change. Wave outcomes classify these changes. For instance:

• Diffusion corresponds to an increase in $H(\Phi\_c)$ for a range of contexts c, often accompanied by increased permeability in the induced transition graph.

• Absorption corresponds to an asymmetric shift where, for many c, out(c) changes from a losing basin to a winning basin and then stabilizes.

• Fusion corresponds to a reduction in the number of peaks returned by A across contexts.

Thus the coupling calculus does not replace wave dynamics; it provides the interface between clause environments and well-internal dynamics.

N.6 A non-clausal-primitive theorem (interface form).

Theorem N1 (Interface sufficiency). If two particles p,q are both particle-origin (Particle_L) and differ only in their boundary-conditioning operators ($B\_p \neq B\_q$), then they can induce distinct clause-level outputs ($\text{out\_p} \neq \text{out\_q}$) without being clause primitives, provided A is sensitive to the differences in Φ conditioning.

Proof sketch. Since $B\_p \neq B\_q$, there exists a context c and state v such that $\Phi\_c(v \mid p) \neq \Phi\_c(v \mid q)$. If A is sensitive to this difference (e.g., it selects argmax and the difference changes which v is maximal), then $\text{out\_p}(c) \neq \text{out\_q}(c)$. Interpretational differences follow if I distinguishes these outputs.

The theorem is simple, but it is the correct structural point: clause-level consequences do not entail clause-level primitive status. They can arise through conditioning and attractor selection at the interface.

Appendix O — Pronouns as particle systems: extended inference and checklist

Pronouns are the paradigmatic case where "clausal relevance" and "non-root status" coincide. Pronouns participate in reference, agreement, and clause-level binding; they therefore appear "syntactic." Yet their inventories are closed, their stems recur across constructions, and their internal morphology is often atomized rather than compositional in the root sense.

This is precisely the signature of particle stratification in MORPHIX: pronominal stems are not lexical roots; they are assembled from a compact particle layer plus compact suffix atoms.

O.1 Why pronouns are expected to be particle-origin in the well.

In the confinement geometry, root material corresponds to stable basins that host rich derivation and open-class expansion. Pronoun systems do not look like that: they are resistant to neologism, they are highly frequent (ψ has strong peaks), and they are portable across clause environments (Φ coupling is high). These are surface/middle properties, not deep root properties.

Moreover, pronoun systems are typically locus-sensitive: small changes in context (person, deixis, discourse prominence) can flip the selected form, which is exactly what boundary conditioning predicts.

O.2 Particle-origin diagnostics specific to pronouns.

The Root_L vs Particle_L audit (Appendix E) can be specialized:

• Head-capable derivation: pronouns rarely host derivational morphology as heads; instead they receive morphology as agreement or case exponents.

• Compounding resistance: pronoun stems generally do not compound productively as lexical items.

• Alternation locality: pronoun alternations are usually limited and structured (suppletion or small paradigms), consistent with sparse E.

• Relational licensing: pronoun selection is conditioned by relational features (person, deixis, discourse), aligning with RelExp_L.

O.3 Checklist for incorporating pronoun evidence into the particle-origin bound-morphology thesis.

Step A: Identify a pronominal subsystem (e.g., demonstratives, relatives, interrogatives).

Step B: Extract the stem inventory and estimate ψ across environments.

Step C: Build the compatibility / licensing matrix across dialects, registers, or branches (analogous to Table 12).
Step D: Audit particle typing for stems and suffix atoms separately.
Step E: Track changes over time; diagnose wave outcomes.
Step F: Check whether bound atoms (case markers, agreement exponents) behave as stabilized particle atoms or as root-driven affixes by comparing their E profiles and their coupling patterns.

O.4 How pronoun evidence supports the IE compact atom thesis.
If pronoun stems are particle-origin, then the bound atoms that recur in pronoun morphology are predicted to be particle-origin as well: they are part of the same portable closed class.
This is the bridge from “pronouns are particles” to “primitive suffix atoms are not clause primitives.”
The bound atoms look like the “deep face” of the same surface inventory: what was once free particle material stabilizes into suffixal atoms through capture and fusion.